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A Multiscale Computational Framework for Structural Organization in Long-Form Chinese Podcast Interviews

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Abstract

Long-form podcast interviews require methods that distinguish episodes, speaker streams, windows, and chronological turns. This study presents a multiscale computational framework for 53 episodes of the Chinese podcast *Yan Zhong Hua Shu* (11,590 turns) and 40 external episodes (20 dialogic, 20 monologic) from four podcasts. Methods combined dependency parsing, Shannon entropy, fixed-token resampling, topic modeling, embeddings, and turn-level lexical similarity. Both roles showed negative mean entropy slopes, while the average host trajectory declined more markedly near the beginning. Negative slopes persisted under 384-token control and alternative window-selection specifications. The paired host minus guest slope contrast was not significant in either the observed or fixed-token analyses. Role-specific embedding–entropy correlations were weak, and turn-level lexical similarity did not increase. Host mean topic divergence tended to exceed guest divergence, but support varied across specifications. Fixed-token document analysis reversed the original entropy ordering between host and guest streams, while external entropy patterns varied across sources. Across the four external sources, the dialogic minus monologic difference in mean dependency distance narrowed from −0.3587 to −0.0661 after sentence-length adjustment. These findings show that declining lexical entropy need not coincide with stronger semantic proximity or local lexical alignment, while external syntactic contrasts require source-aware interpretation.

Keywords: computational discourse analysis; podcast analytics; digital audio; long-form interviews; Shannon entropy; mean dependency distance; sentence embeddings; Jensen–Shannon divergence

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1. Introduction

Podcast platforms have developed into extensive digital archives of long-form audio, transcripts, metadata, and audience activity. Their production ecology is increasingly integrated into contemporary media systems, including Chinese-language platforms and professionally produced independent programs [1–3]. Interview podcasts are especially demanding analytical objects because a single episode may sustain role alternation, autobiographical narration, and recurrent topic movement for an hour or longer [4], while format conventions, distribution practices, and audience engagement continue to evolve [5,6]. Converting these archives into reproducible evidence, therefore, requires methods that can segment large transcript volumes, retain relevant sequence information, and connect each measure to an appropriate unit of analysis.

Podcast research has established a substantial account of audience motivations and listening experience. Convenience, mobility, companionship, authenticity, productivity, and personalized use all shape podcast consumption [7–10]. Transcript-level organization, however, has received less systematic computational attention. Qualitative analysis remains necessary for interpreting meaning and context within individual episodes, whereas comparison across dozens of long interviews requires explicit distinctions among episodes, speaker-specific documents, fixed-length windows, and chronological turns. Large numbers of windows or turns do not increase the number of independent conversations; valid inference, therefore, depends on distinguishing abundant derived segments from the smaller set of episodes and podcasts in which they are nested.

Long-form interviews are produced within an interactional environment shaped by mediated intimacy, politeness, and role performance [11–14]. The host generally controls questioning and framing, whereas the guest often contributes longer stretches of personal or professional narrative, although the linguistic form of both roles varies across episodes, topics, and interview stages. These properties make representation choice consequential. Concatenating a participant's contributions preserves progression within that role's discourse, while chronological representations retain local cross-speaker relations and attribute transitions to the arriving speaker. A scalable workflow must, therefore, preserve role-internal order, shared chronology, episode identity, and source-podcast membership rather than applying all measures to a single undifferentiated transcript.

Extended discourse is organized across several linguistic levels. Conversation analysis describes the sequential organization of interaction [15], information-theoretic research has examined local coordination between dialogue participants [16], and dependency-based measures characterize syntactic configuration and the linear distances associated with grammatical relations [17]. The present framework consequently distinguishes several analytical dimensions. Shannon entropy captures the concentration of unigram lexical probability distributions [18]. Research on entropy-rate constancy examines word predictability conditional on preceding context [19], whereas the unigram entropy used here summarizes lexical probability distributions estimated from token frequencies within each text segment. Embedding cosine similarity describes proximity in semantic vector space; Jensen–Shannon divergence measures movement between topic distributions and mean dependency distance and mean sentence length characterize syntactic organization. Turn-level TF-IDF cosine and Jaccard similarities provide additional measures of surface lexical overlap. Related computational text-analysis frameworks have combined lexical, morphosyntactic, syntactic, and discourse-level features to characterize text complexity and organization [20]. Recent neural topic-modeling research has further integrated sentence embeddings with topic models to enrich semantic representations of textual corpora [21]. The present framework retains these dimensions as complementary but analytically distinct measures.

Kintsch's Construction–Integration model describes how broadly activated information may become progressively constrained during comprehension [22]. Here, it provides a conceptual analogy for discourse development. At the level of speakers' output, recurring words and topic-specific expressions may concentrate lexical probability distributions as an interview develops. Research on audience design, common ground, and local adaptation further indicates that interlocutors coordinate through multiple channels [23–25]. Lexical concentration may, therefore, increase without a corresponding rise in embedding similarity or adjacent-turn lexical overlap, and topic distributions may continue to move even as recurrent lexical choices become more concentrated. Role-separated windows and chronological turns make these possible divergences observable while retaining the episode as the principal inferential unit.

The focal corpus comprises 53 episodes of *Yan Zhong Hua Shu*, a women-focused Chinese podcast centered on personal experience and professional life [26]. The corpus has one recurring host, and role comparisons describe this host's discourse in relation to the sampled guest streams. Research on Chinese women's online discourse situates this orientation within broader practices of identity construction and autobiographical self-positioning [27], while sociolinguistic studies of women-centered interaction provide context for rapport and conversational style [28]. The study does not make a gender comparison; the podcast's orientation instead defines the social and thematic boundaries of the focal dataset. An external corpus of dialogic and monologic podcasts provides a descriptive comparison across the sampled sources beyond this single-program setting.

The central question is how computational measures at different analytical scales characterize the structural organization of long-form podcast discourse. The study makes three contributions. First, it develops a multiscale representation of long-form digital interview transcripts that links episode documents, role-internal windows, chronological turns, and chronological topic windows to defined analytical functions. Second, it evaluates role-internal lexical entropy trajectories using fixed-length windows and window-level fixed-token rarefaction. Third, it compares dialogic and monologic podcast discourse through episode-level aggregation and descriptive source comparisons of sentence-length-adjusted mean dependency distance and fixed-token entropy. The overall design combines trajectory analysis, chronological attribution, continuous JSD analysis, topic-model specification robustness checks, and cross-corpus comparison. It thereby separates archive-level topic redistribution from within-episode topic movement and embedding-based semantic proximity, identifies the scale at which each pattern emerges, and provides a workflow that can be extended to other long-form digital audio archives.

2. Materials and Methods

2.1. Corpora, Transcription, and Preprocessing

The focal dataset comprised 53 long-form interview episodes released between 2023 and 2025, including 18 episodes from 2023, 18 from 2024, and 17 from 2025. The episode-by-role representation contained 1,305,174 Chinese characters and comprised 106 documents, with one host document and one guest document for each episode. The chronological representation contained 11,590 turns and 1,293,690 characters. The host contributed 5243 turns, while guests contributed 6347 turns. Construction of the role-separated trajectory representation retained 1,290,895 characters in 1329 non-overlapping 1000-character windows, with tail windows of at least 200 characters retained. The topic and embedding representation contained 1338 windows constructed from text with spaces preserved, using the same minimum tail length of 200 characters. Spaces were removed before constructing the trajectory windows. These totals correspond to distinct analytical representations and are, therefore, reported separately throughout the article.

The external comparison corpus comprised 20 dialogic interview episodes from *Tianzhen Bu Tianzhen* (D1) and *Zhang Xiaojun: Shangye Fangtan Lu* (D2), and 20 monologic narration episodes from *Ziwo Jinhualun* (M1) and *Tan Liren* (M2). Each source podcast contributed 10 episodes. The dialogic corpus contained 529,395 characters, and the monologic corpus contained 325,945 characters. Host and guest measurements within each dialogic episode were combined using character-count weights before comparison. This aggregation produced one observation per episode and placed the dialogic and monologic materials on the same episode-level basis. Further details on source selection and the release periods of the sampled episodes are provided in Supplementary Section S1 and Table S1 (Panel B).

Episodes in both the focal and external corpora were transcribed using Alibaba's (Hangzhou, China) Fun-ASR model through the Tongyi Tingwu web interface and subsequently checked against the audio. Manual verification focused on proper nouns, domain-specific terminology, speaker boundaries, and unintelligible overlapping speech while preserving the chronological order of the conversation. Transcription accuracy was assessed within the focal corpus by comparing ASR output with the audio-verified transcripts for matched interview content retained for analysis. Character error rate (CER) excluding punctuation was the primary measure; punctuation-aware CER and jieba-tokenized word error rate (WER) were secondary measures. Across the retained focal-corpus content, the pooled punctuation-excluded CER was 0.3535%, with an episode-cluster bootstrap 95% confidence interval of [0.2592%, 0.4612%]. Content excluded during corpus selection was omitted from these comparisons. Filler expressions, non-verbal tags, and repeated pause markers were removed before linguistic analysis. The same cleaning principles were applied across both corpora.

Stanza provided sentence segmentation, part-of-speech annotation within the dependency-parsing pipeline, and dependency parsing [29]. For the sentence-length-adjusted MDD comparison, the focal and external corpora were processed with the same Stanza pipeline. Jieba provided lexical segmentation and the lexical part-of-speech counts used in the lexical analyses [30]. Topic-model inputs were restricted to nouns. The custom topic-model stopword list combined a standard Chinese stopword resource with corpus-specific function words and high-frequency background terms. Character counts provided a reproducible segmentation convention for the Chinese transcripts, while token counts produced by jieba were used in lexical probability calculations and rarefaction.

Counts varied across representations because turn cleaning, role aggregation, whitespace handling, window construction, and tail retention occurred at different stages of data processing. Table 1 summarizes the analytical unit, corpus size, preserved structure, and principal use of each representation. Episodes remained the primary independent units for the focal-corpus statistical analyses, even when the corresponding representations contained hundreds or thousands of windows or turns. External comparisons summarized episode-level measurements by source podcast.

Table 1. Corpus composition and analytical representations.

Representation	Analytical Unit	Corpus Size	Preserved Structure	Main Analysis
Focal episode-by-role representation	Episode-by-role document	53 episodes; 106 documents; 1,305,174 characters	Episode and role	Static role profiles
Chronological turn representation	Speaking turn	11,590 turns; 1,293,690 characters	Shared chronological order	Rarefaction and local alignment
Role-separated trajectory representation	1000-character window	1329 windows; 1,290,895 characters	Role-internal order	Entropy trajectories
Topic and embedding representation	1000-character window	1338 windows	Role-internal topic and semantic sequence	Semantic proximity and topic trajectories
Chronological full-transcript representation	1000-character window	53 episode sequences	Shared chronological order	Chronological topic-transition attribution

Table 1. *Cont.*

Representation	Analytical Unit	Corpus Size	Preserved Structure	Main Analysis
External dialogic corpus	Episode	2 source podcasts; 20 episodes; 529,395 characters	Source podcast and episode	Descriptive cross-mode comparison with source-level summaries
External monologic corpus	Episode	2 source podcasts; 20 episodes; 325,945 characters	Source podcast and episode	Descriptive cross-mode comparison with source-level summaries

Note: Both role-internal window representations used non-overlapping 1000-character windows and retained tail windows containing at least 200 characters. Spaces were removed for the trajectory representation and preserved for the topic and embedding representation. The observed entropy slope analysis additionally applied a separate 500-character end-trimming rule, retaining 1186 windows across 106 episode-by-role series. The 1232 within-role transitions linked consecutive windows within the 106 episode-by-role series.

2.2. Analytical Representations and Computational Workflow

Five analytical representations were derived from the focal transcripts. Episode-by-role documents combined all contributions from one speaker within an episode and supported static role profiles. Role-separated 1000-character windows were constructed from each role's contributions in their original order. The entropy trajectory representation contained 1329 windows constructed after space removal. Role-internal Jensen–Shannon divergence and embedding-based semantic-proximity analyses used a space-preserving representation containing 1338 windows. The 1338 windows formed 106 episode-by-role series containing 1232 within-role transitions. Chronological turns retained the observed speaker sequence and supported rarefaction, entropy-gap analysis, TF–IDF cosine similarity, and Jaccard similarity. Chronological full-transcript windows preserved the shared conversation timeline while providing text blocks of sufficient length for topic-distribution estimation.

Each representation supported a defined set of inferences. Static document measures compared the 53 paired host and guest observations. Slopes calculated from role-internal windows summarized change across each participant's accumulated discourse. Chronological cross-speaker turn pairs described local relations between speakers. Chronological topic windows supported topic-transition attribution according to the host and guest character shares in the arriving window. When repeated segment-level observations originated from the same conversation, they were summarized at the episode or episode-by-role level before inference. Host and guest observations from the same episode retained their pairing in role comparisons.

The external corpus used an episode-level representation. For document-level measures, values were initially calculated for the available role documents or narrator streams. Within dialogic episodes, host and guest document-level values were combined using character-count weights. This procedure produced one observation per episode and placed the dialogic and monologic materials on the same episode-level basis. Podcast identifiers were retained for source-level summaries and leave-one-podcast-out sensitivity analysis.

Figure 1 summarizes the construction of the analytical representations and the granularity associated with their measures.

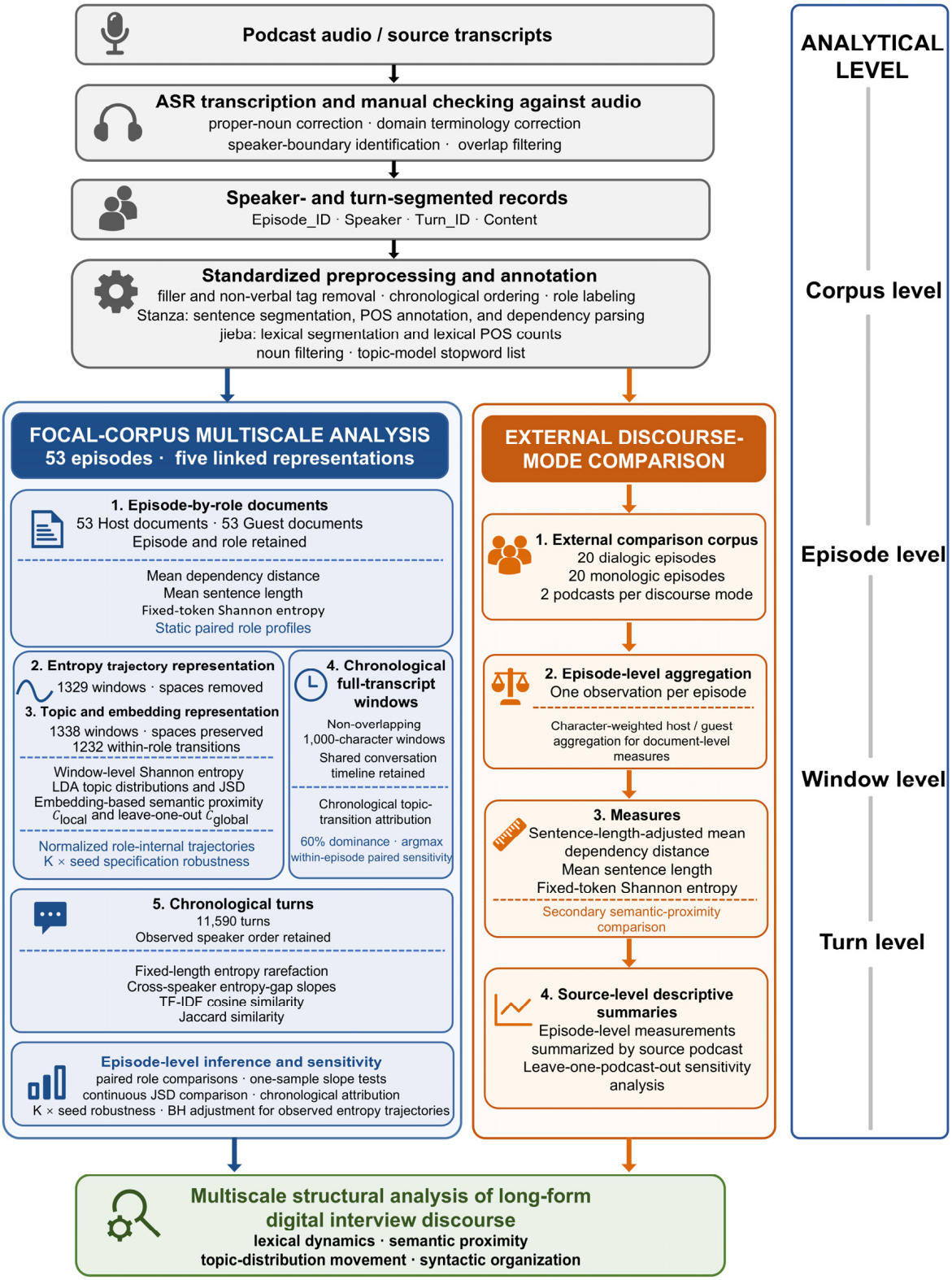


Figure 1. Multiscale computational workflow. Both corpora underwent ASR transcription and manual checking against audio. Focal transcripts were converted into episode-by-role documents, two role-separated window representations, chronological turns, and chronological full-transcript windows. These representations supported static role profiles, role-internal trajectories, local turn analyses, and chronological topic-transition attribution. The external corpus supported source-level descriptive comparisons across the sampled dialogic and monologic podcasts. ASR, automatic speech recognition; POS, part-of-speech; LDA, latent Dirichlet allocation; JSD, Jensen–Shannon divergence; TF–IDF, term frequency–inverse document frequency.

Episode and speaker identifiers were retained throughout the construction of role documents, windows, and turns. Window indices, normalized progress values, and role labels linked the derived observations to their source episodes. Robustness analyses across alternative specifications were kept distinct from the specification adopted for the main descriptive results. This organization supported verification of corpus counts and preserved the distinction between role-internal progress and the shared conversation timeline.

2.3. Linguistic and Computational Measures

Mean dependency distance (MDD) summarized the average absolute linear distance between a dependent and its syntactic head across valid dependency relations [17]. For document d, MDD was calculated as

$$MDD_d = \frac{\sum_{s \in d} \sum_{a \in A_s} |\text{dependent}_a - \text{head}_a|}{\sum_{s \in d} |A_s|} \quad (1)$$

where A_s denotes the set of valid dependency arcs in sentence s. Dependency distance was measured in token positions. Root relations, punctuation dependents, and punctuation heads were excluded. Document-level MDD, therefore, pooled dependency-distance sums and valid-arc counts across all sentences in the document. Mean sentence length was calculated as the average number of Chinese characters per segmented sentence. For the sentence-length-adjusted source analysis, sentence-level MDD was modeled using z_L and z_L^2, where z_L denotes standardized sentence length in Chinese characters. Adjusted estimates were summarized separately by source podcast, with the focal podcast included as a fifth descriptive source.

Shannon entropy quantified concentration within unigram lexical probability distributions [18]. For token probabilities p_i, entropy was calculated as

$$H = -\sum_{i=1}^{n} p_i log_2 p_i \quad (2)$$

where n is the number of observed token types. The probability p_i is the relative frequency of token type i in the analyzed text, and entropy is expressed in bits. Jieba-segmented tokens supplied the frequency distribution. Observed document-level entropy described the complete episode-by-role text, while window-level entropy described consecutive 1000-character segments. Partial windows containing at least 200 characters were retained. The observed trajectory-slope analysis applied a separate 500-character end-trimming rule and retained 1186 windows across 106 episode-by-role series. The full-window-only analysis excluded partial tails and retained 1240 windows across the same series. The remove-first analysis excluded the opening window from each series in the original 1329-window trajectory representation, retaining 1223 windows. It, therefore, used a different window set from the 1186-window end-trimmed analysis. Progress was normalized from 0 to 1 within each analyzed series, and an ordinary least-squares slope summarized change across the normalized progression. Quadratic trajectories were fitted separately to assess departures from linearity.

Turn-level entropy varies with the number of available tokens. The turn-level rarefaction analysis retained turns containing at least 100 segmented tokens and sampled 100 tokens without replacement. Entropy was recalculated for 200 samples from each eligible turn and averaged to obtain H_{rar}. This procedure applied the same token budget to all eligible turns. Correlations between token count and either observed entropy or H_{rar} were compared to assess the reduction in length dependence. The pooled rarefied-turn trajectory was treated as a turn-level length-robustness check because turns remained clustered within episodes.

To assess token-count effects on the entropy trajectories, a separate window-level analysis retained windows containing at least 384 jieba tokens. This analysis included 1272 eligible windows, comprising 1240 full windows and 32 partial tails across all 106 episode-by-role series. Entropy was averaged over 500 samples of 384 tokens drawn without replacement from each eligible window. Ordinary least-squares slopes summarized change in these fixed-token entropy estimates across normalized progress within each episode-by-role series.

Document-level comparisons used separate common budgets of 3095 tokens for the focal role comparison and 5624 tokens for the external comparison. Five hundred samples were drawn without replacement for each eligible document, and their mean entropy provided the fixed-token estimate. These analyses controlled token count at the document level.

The supplementary analysis of verb–adjective balance is described in Supplementary Section S3 [31].

Latent Dirichlet allocation estimated topic-probability distributions from noun-filtered role windows [32]. The adopted descriptive model used $K = 5$ topics, random seed 42, max_iter = 20, max_features = 3000, max_df = 0.95, and min_df = 2. For adjacent topic distributions P and Q, Jensen–Shannon divergence (JSD) was calculated as

$$JSD(P,Q) = 0.5 \times KL(P||M) + 0.5 \times KL(Q||M) \quad (3)$$

where $M = \frac{P+Q}{2}$, and KL denotes Kullback–Leibler divergence. Mean JSD across available within-role transitions summarized topic movement. A secondary topic-transition peak was defined as an adjacent JSD value exceeding the mean of its series by 1.5 standard deviations. Peak ratios were calculated by dividing peak counts by the number of available within-role transitions. The secondary peak analysis used a separate LDA specification from the canonical continuous-JSD analysis.

Chronological topic attribution used the complete sequence of cleaned turns. Non-overlapping 1000-character windows were constructed without separating speakers. The role composition of each arriving window was calculated from its host and guest character counts. A character share of at least 60% defined host-dominant and guest-dominant windows in the primary attribution rule. Windows meeting neither criterion were classified as mixed. An argmax rule assigned every arriving window to the role contributing the larger character share. A within-episode paired specification was also examined for episodes containing both dominance types.

The topic-model specification robustness analysis crossed K values from 3 through 8 with 10 random seeds, producing 60 configurations. Perplexity and normalized pointwise mutual information were used to describe model fit and topic coherence. Host and guest mean JSD values were recalculated within each configuration. The grid assessed the direction and statistical stability of the role contrast across alternative topic spaces. The adopted descriptive results remained based on the $K = 5$, seed = 42 specification. This configuration was retained as a fixed illustrative specification and was not selected by optimizing topic coherence, perplexity, or the significance of the role contrast. The parameter grid was used to assess the sensitivity of the results to topic number and random initialization.

Role-separated windows were encoded using BAAI/bge-small-zh-v1.5 through a local ModelScope installation. The documented chunk-aware specification used a 512-token input limit and produced 512-dimensional vectors. Each window was divided into consecutive, non-overlapping chunks of up to 510 content tokens, without discarding tokens. Chunk-level CLS vectors were L2-normalized and averaged using content-token-count weights, and the resulting window vector was then L2-normalized. Under this tokenizer configuration, 1299 of the 1338 windows (97.09%) exceeded the input limit before

chunking. Local semantic proximity, C_{local}, was calculated as the mean cosine similarity between adjacent window vectors. Global semantic proximity, C_{global}, was calculated as the mean cosine similarity between each window vector and the normalized leave-one-out centroid c_{-t}, calculated from all other window vectors in the same episode-by-role series. Both measures were computed within episode-by-role sequences. The earlier embedding estimates and their descriptive early and late summaries are retained for comparison in Supplementary Section S5.

After turns containing fewer than 50 Chinese characters were removed, the observed entropy-gap analysis used 4500 cross-speaker pairs formed in the filtered sequence. Surface lexical similarity was calculated from 3986 cross-speaker pairs that were strictly adjacent in the original chronological turn sequence, with both turns containing at least 50 Chinese characters. Episode-level regressions estimated change in lexical similarity across normalized chronological progress, whereas observed entropy-gap regressions used chronological pair index.

A fixed-token sensitivity analysis assessed token-count effects on the entropy-gap estimates. Pairs were selected from the length-filtered pair representation when both turns contained at least 100 jieba tokens. For each eligible pair, 100 tokens were sampled without replacement from each turn, and the absolute entropy gap was averaged over 500 draws. Episode-level regressions related these fixed-token gaps to normalized chronological progress.

2.4. Statistical Analysis

The episode served as the primary independent unit for the focal corpus. Static host and guest profiles were analyzed as 53 paired episode-level observations. Temporal analyses estimated one slope for each eligible episode-by-role series. Directional one-sample tests evaluated whether entropy slopes were below zero, and the Host minus Guest difference was tested directly using paired observations. The observed and fixed-token entropy trajectories were analyzed separately. Distributional form was examined with Shapiro–Wilk tests [33], while Wilcoxon signed-rank tests served as nonparametric checks. Effect sizes and 95% confidence intervals were reported for the principal estimates.

The continuous topic-movement analysis compared episode-level mean JSD between paired host and guest streams. The paired contrast was reported with a t-based 95% confidence interval and a separate Wilcoxon signed-rank p-value. Topic-transition peaks were analyzed using a Poisson generalized linear model as a secondary analysis. The logarithm of the number of available transitions entered the model as an exposure offset. Exponentiating the role coefficient yielded the incidence-rate ratio. Dispersion was evaluated using Pearson χ^2 divided by the residual degrees of freedom. Peak ratios were also compared within episodes. Chronological attribution compared transitions associated with host-dominant and guest-dominant arriving windows. Argmax attribution and within-episode paired specifications provided sensitivity checks. The $K \times$ seed grid summarized the direction, effect estimate, and unadjusted p-value for each model configuration as a specification robustness analysis. Role-internal JSD trajectory analyses were reported separately from the paired mean-JSD comparison.

Embedding summaries were calculated across 106 episode-by-role series. Pearson correlations related C_{local} to entropy separately for the 53 host and 53 guest observations. Corresponding correlations involving leave-one-out C_{global} were treated as secondary analyses. Entropy was averaged within each episode-by-role series from the observed entropy windows. Spearman correlations provided sensitivity checks. Paired episode bootstrap resampling retained the host and guest observations from each sampled episode together and estimated 95% confidence intervals for the difference between

role-specific correlations. Turn-level analyses first estimated a progress coefficient for each episode and then tested the 53 episode-level coefficients against zero. For the fixed-token entropy-gap analysis, two-sided one-sample t tests and episode bootstrap confidence intervals summarized the episode-level slopes. This two-stage procedure preserved the episode as the inferential unit. Counts and pooled segment summaries were retained for descriptive reporting.

The external corpus contained 20 dialogic and 20 monologic episode-level observations nested within two source podcasts per discourse mode. External comparisons summarized the sampled sources descriptively. Source-level MDD estimates and the dialogic–monologic contrast were examined before and after sentence-length adjustment. Uncertainty in the adjusted contrast was interpreted conditionally on the sampled sources. Hedges' g was calculated using the pooled standard deviation and the small-sample correction for secondary episode-level comparisons of unadjusted document-level measures. Confidence intervals for these effect sizes were obtained from 5000 bootstrap resamples, with the pooled standard deviation and correction factor recalculated in every resample. These intervals summarized episode-level variability within the sampled source set. Leave-one-podcast-out sensitivity analysis repeated the unadjusted episode-level comparisons after removing one source podcast, producing four sensitivity estimates for each metric.

Benjamini–Hochberg adjustment was applied to three episode-level observed entropy trajectory tests, comprising the directional one-sample tests of host and guest slopes and the two-sided paired Host minus Guest slope contrast. This three-test family replaced the nine-test family used in the previous submission, which combined analyses addressing different research questions and inferential units. The revised family was defined during peer-review revision rather than prospectively, and was restricted to the three related episode-level observed entropy trajectory tests. Wilcoxon tests were retained as nonparametric checks and were not included in the adjustment set. Fixed-token analyses, window-selection sensitivity analyses, role-stratified embedding correlations, topic analyses, and turn-level comparisons were reported separately with their respective effect estimates and, where applicable, unadjusted p-values. These analyses addressed distinct linguistic measures or sensitivity questions, and their unadjusted p-values were interpreted cautiously with regard to multiple testing and analytical choices, irrespective of the direction or statistical significance of the results. The three tests and their BH-adjusted q-values are reported in Supplementary Section S8. Statistical significance for the three adjusted tests was evaluated at $q < 0.05$. Other p-values were interpreted according to their analytical scope and inferential units. Software versions, preprocessing specifications, stopword construction, and additional analytical details are reported in the Supplementary Materials.

3. Results

3.1. Role Profiles and Lexical Entropy Trajectories

Across the 53 episodes, mean sentence length showed the clearest static role contrast. Guest documents averaged 30.840 characters per sentence, compared with 28.421 for host documents. The paired Guest minus Host difference was 2.419 characters, with a 95% confidence interval of [1.925, 2.913] and $p < 0.001$. Mean dependency distance was 3.518 for guests and 3.487 for hosts. The mean difference was 0.031, with a parametric p-value of 0.036 and a Wilcoxon p-value of 0.060. Observed document-level entropy averaged 8.368 for guests and 8.326 for hosts, and the role difference was not significant $p = 0.120$. With a common budget of 3095 tokens, fixed-token entropy was higher for hosts (8.1923 bits) than for guests (8.0698 bits). The paired Guest minus Host difference was -0.1224 bits, with a 95% confidence interval of $[-0.1625, -0.0823]$, an unadjusted p-value of 1.205×10^{-7}, and

$d_z = -0.842$. The complete static profile, including the secondary Busemann's Q measure, is reported in Supplementary Section S4.

Temporal estimates from observed entropy windows showed declining lexical entropy across role-internal progress. Host slopes averaged -0.0775, with a 95% confidence interval of $[-0.1125, -0.0426]$, directional $p = 2.247 \times 10^{-5}$, and Wilcoxon $p = 0.000160$. The standardized one-sample effect size was -0.612. Guest slopes averaged -0.0401, with a 95% confidence interval of $[-0.0721, -0.0082]$, directional $p = 0.007436$, Wilcoxon $p = 0.03399$, and a standardized effect size of -0.346. Both slopes remained significant after Benjamini–Hochberg adjustment across the three episode-level observed entropy trajectory tests, with $q = 0.0000674$ and $q = 0.0112$, respectively.

The direct Host minus Guest slope contrast was -0.0374, with a 95% confidence interval of $[-0.0877, 0.0129]$. The paired parametric p-value was 0.1413, and the Wilcoxon p-value was 0.0759. The paired effect size was $d_z = -0.205$ and the adjusted q-value was 0.1413. The two role streams showed the same directional pattern, although the estimated decline was numerically larger for hosts. Figure 2A shows the normalized trajectories of observed entropy, while Figure 2C presents the observed episode-level slope estimates and their direct contrast. In Figure 2A, the mean host trajectory shows its largest visible decrease near the beginning, followed by smaller fluctuations over the remaining progress. The mean host slope of -0.0775 bits across normalized progress corresponds to approximately 1.1% of the initial plotted entropy level of about 7.29 bits.

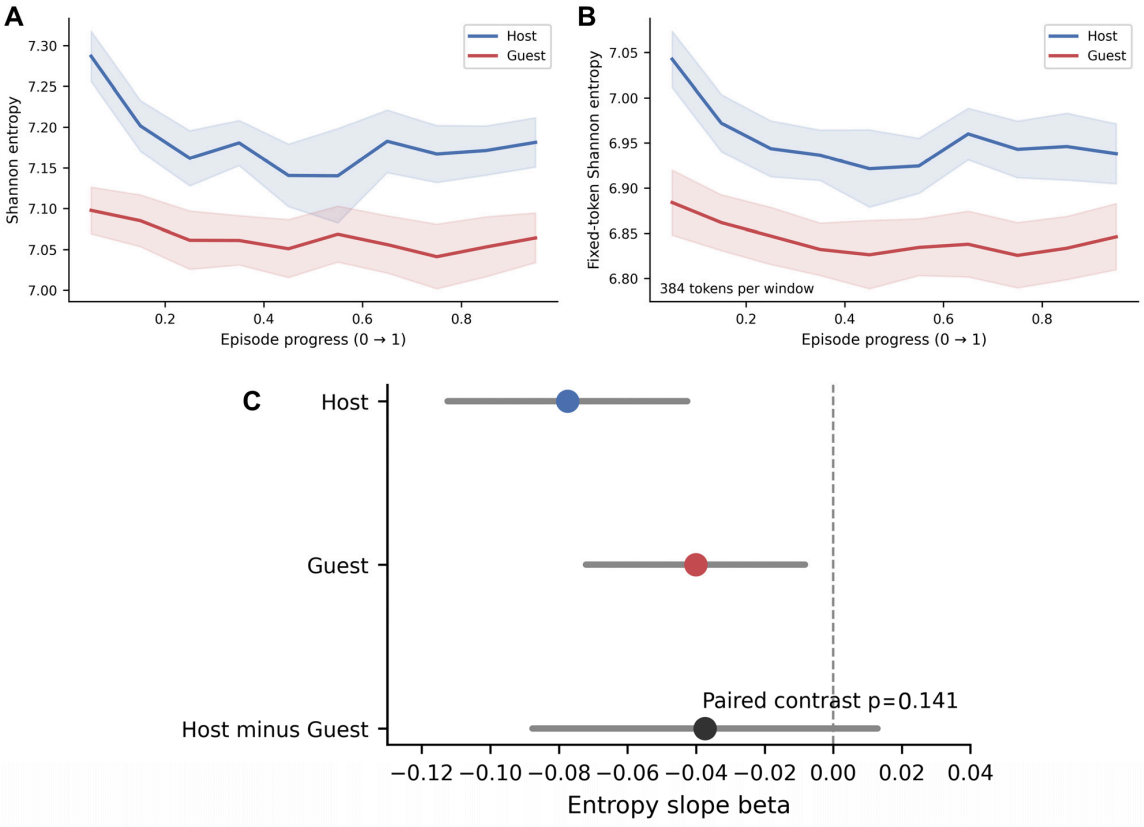


Figure 2. Role-internal lexical entropy trajectories and length control. (**A**) Mean window-level entropy across normalized role-internal progress. Shading indicates 95% confidence intervals. (**B**) Mean fixed-token Shannon entropy across normalized role-internal progress, using 384 tokens per window. (**C**) Observed host and guest episode-level entropy-slope estimates and their paired contrast with 95% confidence intervals. The vertical axes in panels A and B are truncated to make the relatively small absolute changes in entropy visible.

The 384-token analysis retained 1272 windows across all 106 episode-by-role series. Mean entropy slopes remained negative for hosts (−0.0620) and guests (−0.0336). The paired Host minus Guest contrast was −0.0284, with a 95% confidence interval of [−0.0760, 0.0191] and an unadjusted *p*-value of 0.2355 (Table 2). Figure 2B illustrates the fixed-token trajectories. The full-window-only analysis retained 1240 windows and yielded mean slopes of −0.0659 for hosts and −0.0414 for guests, with a non-significant paired contrast of −0.0245 ($p = 0.3027$). The remove-first analysis retained 1223 windows from the original 1329-window representation and yielded mean slopes of −0.2763 for hosts and −0.1243 for guests. The paired contrast increased in magnitude to −0.1520 ($p = 0.000296$). Quadratic fits yielded negative quadratic coefficients in 43 of the 53 host series and 36 of the 53 guest series, with mean coefficients of −0.6329 and −0.2698, respectively. The individual-series fits, therefore, did not uniformly support an early decline followed by stabilization. Detailed sensitivity estimates, early–late comparisons, and turn-level rarefaction results are provided in Supplementary Section S4 and Data S1.

Table 2. Episode-level entropy trajectory estimates.

Analysis	*N*	Mean Slope or Difference	95% CI	*p*	Benjamini–Hochberg *q*
Observed Host entropy slope	53	−0.0775	[−0.1125, −0.0426]	2.247×10^{-5}	0.0000674
Observed Guest entropy slope	53	−0.0401	[−0.0721, −0.0082]	0.007436	0.0112
Observed Host minus Guest entropy slope	53 pairs	−0.0374	[−0.0877, 0.0129]	0.1413	0.1413
Fixed-token Host entropy slope	53	−0.0620	[−0.0967, −0.0274]	0.0003651	—
Fixed-token Guest entropy slope	53	−0.0336	[−0.0624, −0.0048]	0.01152	—
Fixed-token Host minus Guest entropy slope	53 pairs	−0.0284	[−0.0760, 0.0191]	0.2355	—

Note: Observed slope estimates used 1186 windows after 500-character end trimming, whereas fixed-token estimates used 1272 eligible windows sampled at 384 tokens per window. Both specifications retained 106 episode-by-role series from 53 paired episodes. The *p*-values were obtained from directional one-sample tests for role-specific slopes and two-sided paired tests for role contrasts. BH-adjusted *q* values were calculated for the three episode-level observed entropy trajectory tests defined in Section 2.4 and reported in Supplementary Section S8. Fixed-token estimates were reported with unadjusted *p*-values and were not included in this family. Dashes indicate that no *q*-value is reported.

3.2. Semantic Similarity, Topic Dynamics, and Local Alignment

Embedding-based measures showed a different pattern from lexical entropy. Under the documented chunk-aware specification, mean local semantic proximity was 0.7681, and mean leave-one-out global semantic proximity was 0.8362 across the 106 episode-by-role series. The role-stratified correlations with observed window entropy were calculated using this specification. Across the 53 host series, the correlation between C_{local} and entropy was $r = -0.1539$, with a 95% confidence interval of [−0.3919, 0.1100] and $p = 0.2713$. Across the 53 guest series, the correlation was $r = -0.1097$, with a 95% confidence interval of [−0.3235, 0.1225] and $p = 0.4343$. The paired Host minus Guest correlation difference was −0.0442, with a paired episode-bootstrap 95% confidence interval of [−0.3406, 0.2432]. Secondary correlations between leave-one-out C_{global} and entropy were close to zero for hosts $r = -0.0493$ and guests $r = 0.0168$. Further chunk-aware estimates, earlier embedding results, descriptive early and late values, and the external descriptive comparison are provided in Supplementary Section S5.

Topic-distribution analyses showed specification-dependent role contrasts. Under the adopted $K = 5$ and seed 42 specification, mean role-internal JSD was 0.2464 for hosts and 0.2295 for guests. The paired Host minus Guest difference was 0.0169, with a t-based 95% confidence interval of [−0.0076, 0.0415]. The separate paired Wilcoxon test yielded $p = 0.2071$. In the role-internal trajectory analysis, the guest stream had a positive mean

slope of 0.0492, with an unadjusted parametric p-value of 0.0228 and a Wilcoxon p-value of 0.0211. The host mean slope was 0.00254 and was not significant, $p = 0.937$. The episode-averaged trajectory was also not significant, $p = 0.177$, and the direct Host minus Guest slope contrast was not significant, $p = 0.238$.

The secondary transition-exposure Poisson model yielded an incidence-rate ratio of 1.2066, with a 95% confidence interval of [0.8207, 1.7738] and $p = 0.3395$. Peak ratios averaged 0.1054 for guests and 0.0908 for hosts. Their paired comparison was not significant, $p = 0.247$, with $d_z = 0.16$. Chronological attribution provided a separate view of topic movement. Mean JSD was 0.2437 for transitions associated with host-dominant arriving windows and 0.2303 for those associated with guest-dominant arriving windows. The pooled dominance comparison was not significant, $p = 0.3593$. Under the alternative argmax attribution rule, mean JSD was 0.2221 for host-attributed transitions and 0.2284 for guest-attributed transitions, with an unadjusted p-value of 0.9733. The topic-model specification robustness analysis showed variation in the magnitude and inferential support of the role contrast. Host mean JSD exceeded guest mean JSD in 53 of the 60 $K \times$ seed configurations, and 31 configurations had unadjusted p-values below 0.05. The median Host minus Guest mean-JSD difference was 0.0231, with a range from -0.0239 to 0.0651. Further attribution results, within-episode paired comparisons, and specification robustness details are provided in Supplementary Section S6.

Local-alignment analyses provided a complementary chronological perspective. The mean episode-level entropy-gap slope was -0.000165 per chronological pair index (SD = 0.002171). It was not significant according to either the episode-level t-test, $p = 0.582$, or the Wilcoxon test, $p = 0.487$. The analysis included 4500 cross-speaker pairs formed in the length-filtered sequence. The 100-token sensitivity analysis retained 675 pairs in which both turns contained at least 100 jieba tokens, representing 15% of the 4500 length-filtered pairs across all 53 episodes. The mean entropy-gap slope across normalized chronological progress was -0.3118, with a t-based 95% confidence interval of $[-0.9411, 0.3174]$ and a two-sided p-value of 0.3246. The episode-bootstrap 95% confidence interval was $[-0.9612, 0.0355]$. Across the 3986 strictly adjacent cross-speaker pairs, mean TF–IDF cosine similarity was 0.108, and episode-level cosine-similarity slopes were negative on average, $p = 0.00818$. Mean Jaccard similarity was 0.142, and the corresponding slopes did not differ significantly from zero, $p = 0.434$. Local lexical overlap, therefore, did not increase across chronological progress. The negative TF–IDF trajectory was reported with an unadjusted p-value. Complete turn-level results are reported in Supplementary Section S10.

Archive-level topic weights also changed across the three-year period. Topic 4, labeled Life Narratives and Professional Identity, decreased from 0.115 in 2023 to 0.080 in 2024 before increasing to 0.334 in 2025. Topic 2, labeled *Performing Arts and Stage Experience*, decreased from 0.160 to 0.098. Topic 1 increased in 2024 and subsequently declined in 2025; Topic 3 remained comparatively stable, and Topic 5 declined after exhibiting similar weights during the first two years. The five trajectories, therefore, did not follow a common monotonic pattern. Dominant-topic subgroup comparisons for Topics 4 and 2 showed no stable differences in mean dependency distance, mean sentence length, or the supplementary Busemann's Q measure between dominant and non-dominant documents. Within this descriptive model, changes in archive content were more pronounced than changes in the selected structural profiles. Topic labels, keywords, model-selection diagnostics, annual topic weights, trajectory results, and subgroup comparisons are reported in Supplementary Section S9. The main results for semantic proximity, topic-distribution dynamics, and turn-level alignment are summarized in Table 3.

Table 3. Semantic, topic-distribution, and turn-level results.

Analysis	Estimate 1	Estimate 2	Comparison	*p*	Statistical Status
Host C_{local} with entropy	$r = -0.1539$	95% CI [−0.3919, 0.1100]; $N = 53$	Role-specific Pearson correlation	0.2713	Not significant
Guest C_{local} with entropy	$r = -0.1097$	95% CI [−0.3235, 0.1225]; $N = 53$	Role-specific Pearson correlation	0.4343	Not significant
Host minus Guest C_{local}–entropy correlation difference	$\Delta r = -0.0442$	95% CI [−0.3406, 0.2432]	Paired episode bootstrap; 53 pairs	—	CI includes zero
Host leave-one-out C_{global} with entropy	$r = -0.0493$	$N = 53$	Role-specific Pearson correlation	—	Secondary
Guest leave-one-out C_{global} with entropy	$r = 0.0168$	$N = 53$	Role-specific Pearson correlation	—	Secondary
Mean role-internal JSD	Host = 0.2464	Guest = 0.2295	Paired role comparison (Wilcoxon)	0.2071	Not significant
Host role-internal JSD slope	0.00254	$N = 53$	One-sample slope test	0.937	Not significant
Guest role-internal JSD slope	0.0492	$N = 53$	One-sample slope test	0.0228	Positive trajectory; unadjusted
Host minus Guest JSD-slope contrast	Host and Guest slopes	$N = 53$ pairs	Direct paired contrast	0.238	Not significant
Exposure-adjusted peak rate	IRR = 1.2066	95% CI [0.8207, 1.7738]	Role coefficient	0.3395	Not significant; secondary
Peak ratio	Host = 0.0908	Guest = 0.1054	Paired role comparison	0.247	Not significant
Chronological JSD	Host dominant = 0.2437	Guest dominant = 0.2303	Pooled dominance comparison	0.3593	Not significant
$K\times$ seed mean-JSD contrasts	Median $\Delta = 0.0231$	Range [−0.0239, 0.0651]	Host > Guest in 53/60 configurations; 31/60 raw $p < 0.05$	—	Specification robustness
Observed entropy-gap slope	−0.000165	SD = 0.002171	Episode-level slope mean per chronological pair index	0.582	Not significant
Fixed-token entropy-gap slope	−0.3118	95% CI [−0.9411, 0.3174]	Episode-level *t*-test across normalized progress; $N = 53$	0.3246	Sensitivity; eligible subset
TF-IDF cosine	Mean = 0.108	3986 turn pairs	Episode-level slope test	0.00818	Exploratory decline
Jaccard	Mean = 0.142	3986 turn pairs	Episode-level slope test	0.434	Not significant

Note: All *p*-values shown are unadjusted. The paired Host minus Guest mean-JSD difference was 0.0169, with a t-based 95% confidence interval of [−0.0076, 0.0415]; the corresponding *p*-value was obtained from a separate Wilcoxon signed-rank test. Correlation-difference intervals were obtained by paired episode bootstrap. The fixed-token entropy-gap analysis used 100 tokens per turn and retained 675 of 4500 pairs across all 53 episodes, representing 15% of the length-filtered pair set. Its reported confidence interval is t-based. Dashes indicate that no *p*-value is reported.

3.3. External Discourse-Mode Comparison

Across the two dialogic sources (D1 and D2) and two monologic sources (M1 and M2), the unadjusted dialogic minus monologic MDD contrast was −0.3587. After sentence-length adjustment, the contrast was −0.0661, with a 95% confidence interval of [−0.1006, −0.0315] conditional on the four sampled sources (Figure 3C; Table 4, Panel B).

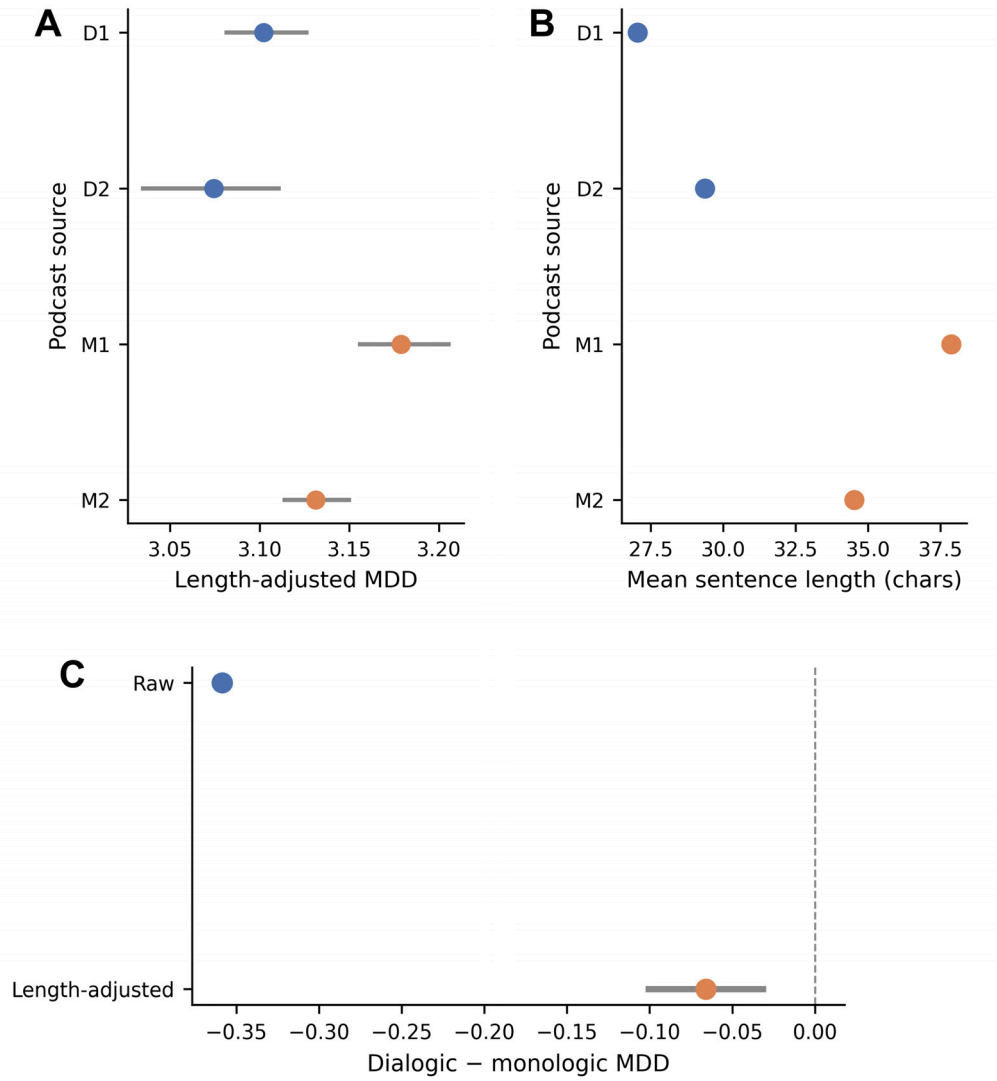


Figure 3. Source-level external comparison of dialogic and monologic discourse. (**A**) Source-level sentence-length-adjusted MDD with source-specific confidence intervals. (**B**) Source-level mean sentence length. (**C**) Dialogic minus monologic MDD contrasts before and after sentence-length adjustment. The interval for the adjusted contrast is a conditional 95% confidence interval. Negative values indicate lower MDD in dialogic sources. D1 and D2 denote dialogic sources, and M1 and M2 denote monologic sources. In panels A and B, blue dots denote dialogic sources (D1 and D2), and orange dots denote monologic sources (M1 and M2). In panel C, the blue dot represents the unadjusted contrast, and the orange dot represents the sentence-length-adjusted contrast.

Both dialogic sources had shorter mean sentences than either monologic source (Figure 3B). Sentence-length-adjusted MDD was 3.1022 for D1 and 3.0744 for D2, compared with 3.1789 for M1 and 3.1313 for M2 (Figure 3A). The range across the four external sources was 0.1045, exceeding the absolute adjusted dialogic–monologic contrast of 0.0661. The focal podcast had an adjusted MDD of 3.1485, with a 95% cluster-bootstrap confidence interval of [3.1389, 3.1582]. The focal-podcast estimate fell between the adjusted MDD estimates of the two monologic sources.

At the common budget of 5624 tokens, fixed-token entropy averaged 8.3558 in dialogic episodes and 8.2946 in monologic episodes, with a secondary episode-level Hedges' g of 0.328. Observed document-level entropy averaged 8.373 in dialogic episodes and 8.413 in monologic episodes. The descriptive ordering, therefore, reversed after token control, and source-level fixed-token entropy estimates did not show a consistent dialogic–monologic pattern. In leave-one-podcast-out analyses of unadjusted document-level measures, the

dialogic sources retained lower mean dependency distance and mean sentence length across all four source omissions, whereas the observed Shannon entropy contrast changed direction depending on the omitted source (Table S9). Complete source-level and episode-level descriptive results for structural measures are reported in Supplementary Section S7, and external embedding estimates in Section S5.

Table 4. Source-level MDD estimates and external fixed-token entropy summary.

Panel A. Source-Level Sentence-Length-Adjusted MDD		
Source	**Source type**	**Adjusted MDD**
D1	Dialogic	3.1022
D2	Dialogic	3.0744
M1	Monologic	3.1789
M2	Monologic	3.1313
Focal podcast	Descriptive reference	3.1485
Panel B. External Comparison Summary		
Analysis		Result
Unadjusted MDD contrast, dialogic minus monologic		−0.3587
Sentence-length-adjusted MDD contrast, dialogic minus monologic		−0.0661, 95% CI [−0.1006, −0.0315]
Fixed-token Shannon entropy		Dialogic = 8.3558 bits; Monologic = 8.2946 bits
Secondary fixed-token entropy effect size		Hedges' g = 0.328

Note: D1 and D2 denote dialogic sources, and M1 and M2 denote monologic sources. Sentence-length-adjusted MDD accounts for the linear and quadratic association between sentence length and sentence-level MDD. The adjusted dialogic minus monologic contrast and its confidence interval are conditional on the four sampled external sources. The focal podcast is included as a fifth descriptive source. Fixed-token entropy used a common budget of 5624 tokens per episode. Source-specific confidence intervals are shown in Figure 3A, and the focal-podcast confidence interval is reported in the text.

4. Discussion

4.1. Structural Organization Across Analytical Scales

Comparison across analytical levels provides a differentiated account of structural organization in long-form podcast interviews. Both role streams showed negative mean lexical entropy slopes within episodes. Negative mean slopes also persisted under fixed-token control and alternative window-selection specifications. Embedding-based semantic proximity was only weakly associated with entropy, and mean role-internal JSD tended to be higher for hosts. The magnitude and inferential support of the JSD contrast varied across topic-model specifications. The external MDD contrast narrowed substantially after sentence-length adjustment. The range of source-level adjusted MDD estimates exceeded the absolute adjusted contrast. These results locate the most consistent within-episode pattern in lexical distributions while placing the external syntactic contrast within substantial source-level variation.

The multiscale framework is central to this interpretation. Episode-by-role documents describe static profiles of accumulated host and guest language. Role-internal windows support the estimation of progression within each speaker stream. Chronological turns and full-transcript windows retain the shared sequence required for cross-speaker alignment and topic-transition attribution. External episode-level observations were summarized within source podcasts, while podcast identity was retained for sensitivity analysis. Source-level estimates characterize differences among the sampled dialogic and monologic podcasts. Although these representations derive from the same general class of transcript data, each preserves a different aspect of the original discourse structure.

The analytical measures likewise describe complementary dimensions of organization. Their partial divergence indicates the scale at which each pattern emerges. Entropy may decline as recurring lexical choices concentrate probability mass, even when adjacent embedding vectors do not become increasingly similar. Topic distributions may shift among themes constructed from noun co-occurrence while surface lexical overlap remains limited. Mean sentence length and mean dependency distance describe the grammatical configuration of produced sentences. In the external corpus, sentence-length adjustment reduced the MDD contrast, whereas token-count control reversed the entropy ordering. Research on dependency locality and referential choice provides a context for interpreting the syntactic dimension [34–37], while Shannon's information-theoretic formulation provides the direct basis for interpreting concentration within lexical probability distributions [18].

This representation-aware approach contributes to digital media analytics by linking large transcript archives to reproducible analytical units. Segmentation and aggregation form substantive components of the research design. The framework also provides a practical basis for length control and evaluation of analytical choices. Fixed-token analyses control token count in entropy estimation, while the full-window-only and remove-first sensitivity analyses assess the dependence of trajectories on window inclusion. Chronological windows preserve the shared conversation timeline. The topic-model specification robustness analysis assesses variation across alternative topic spaces, and leave-one-podcast-out sensitivity analysis evaluates how particular source podcasts affect secondary episode-level summaries. These procedures make long-form digital audio archives accessible to computational comparison while retaining the nested structure of episodes, speakers, windows, turns, and source podcasts.

4.2. Lexical Concentration, Semantic Proximity, and Topic Dynamics

The decline in entropy across both role streams indicates changes in lexical probability distributions, although the mean host trajectory showed a more pronounced early decrease followed by relatively small fluctuations. The negative mean slopes persisted under fixed-token control, but they summarize change across normalized progress and do not imply a uniform decline throughout each interview. The direct Host minus Guest slope contrast remained non-significant in the observed and fixed-token analyses. The remove-first sensitivity analysis yielded a larger role contrast, but it used a different window set from the primary end-trimmed analysis. The increased contrast, therefore, cannot be attributed to the removal of opening windows alone. The principal conclusion concerns the negative mean slopes observed for both role streams, without establishing a stable difference between them. The document-level entropy ordering reversed under token-count control, indicating that static role differences were sensitive to text length. Kintsch's account of progressive constraint during comprehension provides a conceptual analogy rather than a direct explanation of the observed production patterns [22]. The corpus thus provides evidence of changes in unigram lexical distributions without identifying a specific cognitive mechanism.

Long-form interviews in this corpus frequently develop personal and professional narratives. Research on autobiographical organization has described how recurring events and self-relevant themes structure extended accounts [38,39]. Studies of institutional interaction clarify how questioning rights and knowledge displays are distributed across participant roles [40,41]. Narrative-positioning research further connects local acts of telling with identities constructed across extended discourse [42–44]. Recurrent names, experiences, and interpretive frames may, therefore, contribute to the lexical concentration observed in later role-internal windows. Within the adopted topic model, the mean weight of the topic labeled Life Narratives and Professional Identity was higher in 2025

than in 2023, despite a decline in 2024. This diachronic redistribution describes changes in archive composition, whereas the entropy trajectories describe progression within individual episodes.

Embedding-based results refine this account. Role-stratified associations with entropy were weak for C_{local} and close to zero for leave-one-out C_{global}. Under the earlier embedding specification, descriptive early and late C_{local} values showed only a limited difference. Embedding-based semantic proximity and lexical concentration describe different properties of the transcripts. Chronological turn analyses provided a similar qualification. Entropy gaps did not narrow reliably in the length-filtered turn analysis, while TF–IDF cosine similarity was low and showed a negative unadjusted trajectory, and Jaccard similarity showed no significant trajectory. The fixed-token entropy-gap analysis also yielded no significant mean trend, but retained only the 15% of pairs in which both turns contained at least 100 jieba tokens. This result, therefore, applies to the eligible longer exchanges and cannot be generalized to all adjacent turns. Changes in role-internal lexical entropy did not entail increased lexical overlap between adjacent cross-speaker turns.

Topic-distribution movement produced specification-dependent role contrasts. Mean role-internal JSD did not differ significantly between hosts and guests under the adopted model. The guest trajectory showed a positive mean slope with an unadjusted p-value below 0.05, whereas the host trajectory, episode-averaged trajectory, and direct Host minus Guest contrast were not significant. This result, like the negative TF–IDF cosine trajectory, was interpreted cautiously as exploratory rather than confirmatory evidence. Because peak thresholds were defined relative to each series' own JSD distribution, the peak analysis remains secondary to the continuous-JSD comparison. Peak rates adjusted for available transitions and paired peak ratios showed no significant role difference. Under the 60% dominance rule, chronological attribution yielded numerically higher mean JSD for host-dominant arriving windows, although the pooled comparison was not significant and the ordering reversed under argmax attribution. Host mean JSD exceeded guest mean JSD in most configurations of the topic-model specification robustness analysis. The effect magnitude and inferential support varied across alternative topic spaces. Research on institutional interaction and epistemic roles remains relevant to understanding how questions, knowledge displays, and topic rights are organized [40,41]. In the present corpus, inferential support for topic-based role contrasts was less stable than the shared entropy trajectory.

4.3. Syntactic Variation Across Dialogic and Monologic Sources

Sentence length accounted for much of the unadjusted MDD contrast between the sampled dialogic and monologic sources. Both external dialogic sources had shorter mean sentences and lower sentence-length-adjusted MDD than either monologic source. The adjusted MDD contrast retained the same direction but was substantially smaller. Source-podcast identity, topic, speaker composition, scripting, editing, and production style remain relevant to the interpretation of this cross-corpus contrast.

Research on dependency locality provides a basis for interpreting shorter dependency relations as a distinct configuration of grammatical material [34–36]. Narrative research also indicates that referential and syntactic choices vary with discourse task and production context [37]. The observed differences among the sampled sources may motivate hypotheses about dialogic and monologic production conditions, but the present comparison cannot establish these mechanisms.

One hypothesis for future research is that turn-taking may facilitate incremental planning, feedback, and floor transfer. Interview participants may distribute information across multiple contributions, whereas monologic narrators may organize longer stretches

of content within a continuous discourse stream. Whether these production conditions contribute to differences in sentence length and MDD requires further investigation across more independent podcast sources in each discourse mode. The present evidence concerns the structural properties of the transcripts. Prosody, editing, preparation, and delivery style may also contribute to the observed pattern.

Source-podcast identity remains an important boundary of the comparison. The external corpus contains two podcasts within each discourse mode. With only two external sources per discourse mode, the comparison cannot fully separate discourse-mode differences from source-specific variation. The range of source-level adjusted MDD estimates exceeded the absolute adjusted dialogic minus monologic contrast. The focal podcast's adjusted MDD was higher than that of either external dialogic source and fell between the two monologic-source estimates. Its position further illustrates variation among the sampled dialogic sources.

Lexical entropy was sensitive to document length and source composition. Fixed-token control reversed the dialogic and monologic ordering, while source-level entropy estimates did not show a consistent pattern across modes.

4.4. Digital Media Analytics, Limitations, and Future Research

This study provides a scalable workflow for analyzing transcripts from long-form digital audio. Its methodological contribution lies in representation-aware modeling and the alignment of analytical representations, inferential units, and statistical procedures. Transcript processing retains episode, role, window, and turn identifiers. Fixed-length windows create comparable segments for trajectory analysis, while episode-level slope estimation preserves the episode as the inferential unit. Fixed-token analyses address token-count dependence, and transition-exposure adjustment accounts for unequal numbers of available transitions. Sensitivity analyses assess window selection, chronological attribution, and source-podcast influence, while topic-model specification robustness analysis examines variation across topic solutions. Together, these elements support computational comparison across a large archive without treating every derived segment as an independent conversation. Recent text-mining applications have similarly integrated corpus segmentation, topic modeling, and visualization into unified analytical workflows [45]. Comparable media-monitoring systems have also combined sentence embeddings, topic classification, and clustering to organize evolving multilingual media content [46]. The workflow could support researchers and media analysts in comparing discourse structure across episodes and sources. Episode-level summaries could help podcast producers review episode organization, and topic distributions could inform archive indexing and content navigation on digital platforms.

The focal corpus also defines the boundaries of the substantive claims. All 53 episodes were drawn from one women-focused podcast with a recurring host. Host–guest contrasts may, therefore, reflect both participant roles and the recurring host's individual style. The podcast emphasizes lived experience and professional narratives, which may shape lexical recurrence, semantic organization, and topic composition. The corpus contains no gender comparison and represents a predominantly cooperative interview context. Broader generalization will require additional hosts, production teams, interactional genres, and sociocultural settings.

The external corpus extends the design across four podcasts, but source-podcast identity and discourse mode remain partially confounded. Twenty episodes per mode provide episode-level observations nested within two source podcasts per mode. Interpretation of the external contrast remains conditional on these four sampled sources, with limited information about between-podcast heterogeneity. Whether the monologic episodes

followed prepared scripts was not independently verified. The focal podcast serves as an additional descriptive source. Future corpora should sample more podcasts within each discourse mode and retain nested identifiers for episodes, speakers, seasons, and production teams. Multilevel models could then distinguish mode-related patterns from source-specific conventions.

Transcript-based analysis captures lexical, syntactic, semantic, and topic-distribution structure while leaving other communicative channels outside the present measurements. Prosody, pauses, overlapping speech, gesture, editing, music, and audience reception all contribute to the podcast experience. Automatic transcripts were manually checked against the audio, and CER and WER quantified discrepancies between ASR output and audio-verified focal transcripts for matched retained content. A separate quantitative CER assessment was not conducted for the external corpus. Future research could integrate the present transcript representations with acoustic timing, overlap annotation, video, platform metadata, and audience-response data.

Model specification introduces additional constraints. LDA topic labels depend on noun filtering, stopword selection, the number of topics, and random initialization. The topic-model specification robustness analysis showed that role contrasts varied in magnitude and inferential support across alternative topic solutions; topic-derived results should, therefore, remain exploratory or descriptive. Role-separated windows preserve the internal order of each speaker's contributions, whereas chronological representations are required for claims involving the shared conversation timeline. The embedding model produced 512-dimensional vectors. Embedding results remain dependent on model specification and input handling, and window-level representations may obscure finer local discourse changes. The focal-corpus embedding estimates emphasized in Section 3.2 use the documented chunk-aware specification, while the earlier estimates, whose exact historical model version and input handling could not be independently verified, are retained for comparison in Supplementary Section S5. Busemann's Q depends on lexical part-of-speech annotation and was retained only as a secondary descriptive stylistic measure. Future studies could compare alternative window sizes, contextual topic models, longer-context embedding models, and multimodal representations while maintaining episode-level inference.

5. Conclusions

This study developed a multiscale computational framework for analyzing structural organization in long-form Chinese podcast interviews. The framework links episode-by-role documents, role-internal windows, chronological turns, chronological topic windows, and external episode-level observations nested within source podcasts to measures suited to each representation. Within the focal corpus, mean role-internal Shannon entropy slopes were negative for both hosts and guests, with the average host trajectory showing a more pronounced early decrease. The decline persisted under fixed-token control, while the direct difference between host and guest slopes was not significant in the observed and fixed-token analyses. The estimated role contrast was sensitive to window selection. Embedding-based semantic proximity showed only weak associations with entropy, and chronological turn-level measures showed no increase in local lexical overlap. Topic-based role contrasts varied in magnitude and inferential support across topic-model specifications.

The external comparison showed that sentence-length adjustment substantially reduced the unadjusted MDD contrast. Both sampled dialogic sources had shorter mean sentences and lower sentence-length-adjusted MDD than either monologic source, but the range across sources exceeded the adjusted cross-mode contrast. Interpretation of discourse-mode differences, therefore, remains conditional on these four sources. Fixed-

token control reversed the external entropy ordering, and source-level estimates showed no consistent pattern across modes.

The framework provides a transparent and scalable basis for computational analysis of extended digital audio archives. It clarifies how conclusions depend on the preservation of episode, role, window, turn, and source-podcast structure. More broadly, it distinguishes abundant derived segments from the underlying units of inference, a distinction that is essential for statistically valid analysis of large transcript collections.

Supplementary Materials: The following supporting information can be downloaded at: https://www.mdpi.com/article/10.3390/app162010435/s1. The supplementary PDF contains Sections S1–S10, including the following figures and tables. Figure S1: Selected role-sensitive structural profiles in the focal corpus; Figure S2: Embedding-based semantic-proximity summaries; Figure S3: Descriptive host and guest JSD trajectories across normalized role-internal progress; Figure S4: Chronological topic-transition attribution; Figure S5: Topic-model specification robustness across 60 configurations; Figure S6: Annual mean weights of the five descriptive archive topics. Table S1: Corpus composition and analytical representations; Table S2: Software and model specifications for selected analytical procedures; Table S3: Static episode-by-role profiles; Table S4a: Role-internal entropy trajectories; Table S4b: Early–late entropy comparisons and trajectory sensitivity analyses; Table S4c: Turn-level rarefaction-based length check; Table S5a: Focal embedding-based semantic-proximity summaries; Table S5b: External embedding-based semantic-proximity descriptive comparison; Table S6: Topic-transition estimates across role-internal and chronological representations; Table S7: Topic-model specification robustness summarized by K; Table S8a: External structural and lexical descriptive summaries and sentence-length-adjusted MDD; Table S8b: External fixed-token entropy summaries; Table S9: Leave-one-podcast-out results for unadjusted document-level external measures; Table S10: Benjamini–Hochberg adjustment for three episode-level observed entropy trajectory tests; Table S11: Descriptive labels and highest-weighted keywords in the five-topic archive model; Table S12: Annual mean weights in the five-topic archive model; Table S13: Structural comparison of Topic 4-dominant and other documents; Table S14: Structural comparison of Topic 2-dominant and other documents; Table S15a: Episode-level entropy-gap summaries; Table S15b: Surface lexical similarity in the focal corpus. Data S1: Analysis-ready derived data supporting the main results (Excel file); Data S2: Numeric robustness and sensitivity outputs (Excel file).

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Data Availability Statement: Analysis code, parameter files, the topic-model stopword list, and software documentation are available at [https://github.com/wyqmath/podcast-analysis] (accessed on 9 October 2026). Raw podcast transcripts are not redistributed because they remain subject to third-party copyright and platform restrictions.

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Abbreviations

The following abbreviations are used in this manuscript:

ASR	automatic speech recognition
CER	character error rate
WER	word error rate
MDD	mean dependency distance
POS	part-of-speech
LDA	latent Dirichlet allocation
JSD	Jensen–Shannon divergence
KL	Kullback–Leibler divergence
TF–IDF	term frequency–inverse document frequency
CI	confidence interval
IRR	incidence-rate ratio

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